2026 MIDDLE SCHOOL STEM CAMPS

From Trees to Tech

STUDENT HANDOUT PACKET

A week of team STEM missions. Predict, build, test, score, redesign, defend.

IN THIS PACKET

ACTIVITY TTT-01 · BIOLOGY + ENERGY

MudWatt Bioelectric League

Can your team make mud generate electricity and defend your design with data?

Win condition: Highest combined score from peak reading, daily data quality, controlled-variable design, redesign notes, and final explanation.

THE SCIENCE

Electrogenic bacteria. Some bacteria already in ordinary soil are electrogenic: as they digest nutrients, they push spare electrons onto surfaces outside their cells. Bury a carbon-felt anode in oxygen-free mud and these microbes coat it, feeding it a steady stream of electrons.

Completing the circuit. Electrons leave the anode, run through the external wire and a 100 kΩ resistor, and reach the cathode resting in the air on top. There they join oxygen and protons from the mud to form water. A multimeter reads the voltage across the resistor, which by Ohm's law tracks the current the microbes produce. More food and more healthy microbes mean more electrons per second, so the measured voltage rises. Power is roughly voltage times current.

BUILD AND RUN IT

01 **Inoculate the mud.** Gloves on, working over the trash-bag liner. Pack the clear 24 oz deli container with sponge-damp pond mud or soil. The bacteria you need are already in the dirt, so no culture is required.

02 **Place the electrodes.** Bury one 4×4 in carbon-felt square near the bottom as the anode and smooth mud over it so no air reaches it. Rest the second 4×4 in carbon-felt square flat on top as the cathode, exposed to the air. Keep the two squares from touching, and run an alligator lead off each.

03 **Add a measured fuel.** Mix in one measured sugar cube (or another approved fuel in a measured amount). Record exactly how much. This is the food that drives the electron output.

04 **Connect and read a baseline.** Wire the 100 kΩ resistor in line between the electrodes. Set the multimeter to DC millivolts and read the voltage across the resistor (red probe to the cathode side, black to the anode side). Log the first reading. Day one is often only tens of millivolts; the biofilm needs days to grow.

05 **Control one variable.** Pick ONE thing to test all week: fuel type, fuel amount, moisture, or anode packing. Change only that, predict the effect, and compare to baseline.

Safety: Gloves required and build over the trash-bag liner. No food or drink near the mud bench. Keep the two carbon-felt electrodes from touching. Wash hands after handling soil. No goggles needed; the cell is only a few hundred millivolts. Allergy: Do not use nut-containing food scraps. The standard fuel is a measured sugar cube; banana peel, stale bread, or leaf litter are alternatives.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Peak voltage reading	35
Daily data log quality	20
Controlled-variable design	15
Redesign or maintenance plan	15
Final explanation	15
Total	**100**

ACTIVITY TTT-02 · FIELD SCIENCE

Forest Sensor Sprint

Find the smartest place to install a future campus tree sensor.

Win condition: Best evidence-based sensor recommendation with accurate measurements and efficient route planning.

THE SCIENCE

Microclimate varies in meters. Temperature, humidity, light, and soil moisture can change sharply over short distances. A spot under a dense canopy can be several degrees cooler and far more humid than open lawn a few steps away. Good sensor placement starts with measuring, not guessing.

Evidence-based siting. A useful sensor sits where its readings represent the area you care about and where it will not be disturbed. Teams collect data at field stations, then argue for one location using their own numbers plus an efficient route between checkpoints.

BUILD AND RUN IT

01 **Calibrate together.** Take one reading as a whole group at a reference spot so every team's tools agree before they split up.

02 **Run the field stations.** Rotate through stations measuring temperature, humidity, light, and soil moisture. Record units every time.

03 **Estimate tree height.** Sight the treetop through the clinometer straw, read the angle off the plumb string, and use the standoff distance the instructor pre-marked with the long tape: tree height = eye height + distance x tan(angle).

04 **Plan an efficient route.** Order your checkpoints to minimize backtracking. Note the route on the card.

05 **Recommend one site.** Pick one sensor location and defend it with your measurements, not opinion.

Safety: Buddy system outdoors. Stay on approved route. Sun, hydration, and weather check before launch. Allergy: Pollen sensitivity note. Students do not need to touch plants.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Data accuracy	30
Route efficiency	20
Site recommendation evidence	30
Teamwork and field conduct	20
Total	**100**

ACTIVITY TTT-03 · BIOMIMICRY

Seed Dispersal Derby

Build a seed that escapes the parent tree without engines or throwing.

Win condition: Best combined distance, hang time, target landing, biomimicry explanation, and redesign improvement.

THE SCIENCE

Drag, lift, and hang time. A maple samara spins as it falls. The spinning wing pushes against the air and makes lift, so the seed drifts down slowly and even a light breeze carries it away from the shade of its parent. More surface area for the same mass usually means a longer, slower fall.

One variable at a time. Wing length, wing angle, and added mass all change flight. Engineers learn what each does by changing only one and comparing. Copying a natural strategy on purpose is biomimicry.

BUILD AND RUN IT

01 **Study a real samara.** Drop a maple seed or a model. Watch how the wing makes it spin and slow down.

02 **Build a baseline seed.** Make a simple winged seed with one paper clip for mass. Record a baseline drop time and distance.

03 **Change one variable.** Adjust wing length OR mass OR fold angle, only one. Predict the effect before testing.

04 **Test for hang time and distance.** Drop from the standard height or release in the fan lane. Time three trials and average.

05 **Aim for a target.** In a final round, try to land in a marked zone. Control plus distance both score.

Safety: No throwing at people. Use marked launch lanes and low-power fan settings. Allergy: None.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Distance traveled	35
Hang time	25
Target landing	15
Biomimicry explanation	15
Redesign improvement	10
Total	**100**

ACTIVITY TTT-04 · BIOMIMICRY

Xylem Pipeline Relay

Move water uphill like a tree, faster and cleaner than rival teams.

Win condition: Most water delivered with speed, low spill, and strong explanation.

THE SCIENCE

Capillary action. Water climbs narrow spaces on its own. Water molecules stick to a surface (adhesion) and to each other (cohesion), so a thin column is pulled upward against gravity. Narrower channels and more wettable materials pull water higher and faster. Trees use this in their narrow xylem, but on its own it lifts water only about a meter. To reach the treetop, water evaporating from the leaves (transpiration) pulls the whole column up, like sipping through a straw.

Material and geometry. A folded paper towel, a cotton ball pulled into a strand, and a strip of cotton cloth each move water at a different rate. Capillary action only works in fine, connected channels that water can wet, so pre-wet your wick, keep it in firm contact, and use a gentle, short climb. Path length, slope, and contact all matter. Your team picks the material and route to deliver the most water to a target cup.

BUILD AND RUN IT

01 **Test three wick materials.** Pre-wet each wick, then dip the paper towel, the cotton-ball strand, and the cotton cloth into colored water. Watch which climbs fastest and highest.

02 **Pick a material and route.** Choose your wick and lay out a path from the source cup up over a block to the target cup.

03 **Run the first transfer.** Start the wick and time how much water reaches the target in the time limit. Catch spills on the bag-covered table.

04 **Reduce spill and length.** Trim the path, improve contact, and re-run. Less spill and a shorter path usually deliver more.

05 **Final delivery round.** Run the best design for score. Water delivered, speed, and low spill all count.

Safety: Work over the bag-covered table and wipe spills immediately so the floor stays dry. No standing on chairs or stacked furniture to build height. Use washable food coloring; it can stain skin and clothes, so wear gloves if you do not want stained hands. No goggles are needed for moving water.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Water delivered	40
Speed	20
Least spill	15
Design explanation	15
Redesign gain	10
Total	**100**

ACTIVITY TTT-05 · ECOLOGY

Greenhouse Climate Controller

Run the greenhouse controls for a plant that cannot complain, only wilt.

Win condition: Best plant-zone-control match supported by tour evidence.

THE SCIENCE

Controlled environments. A greenhouse lets growers set temperature, humidity, and light to match what a plant needs. Get it wrong and the plant wilts, scorches, or molds. Every setting is a tradeoff: more light can mean more heat and faster drying.

Evidence from the tour. On the greenhouse tour you gather clues about real plant needs and real zone settings. Back at the table you match plant cards to climate dials using that evidence, then defend the layout.

BUILD AND RUN IT

01 **Gather tour evidence.** During the greenhouse visit, note real temperature, humidity, and light clues for different zones.

02 **Read the plant cards.** Each plant card lists needs and tolerances. Sort cards by what they require.

03 **Set the climate dials.** Match each plant to a zone and set temperature, humidity, and light on the dial board.

04 **Resolve tradeoffs.** When two plants want different things in one zone, choose the setting your evidence best supports.

05 **Defend the layout.** Explain each placement using a specific clue from the tour.

Safety: Stay with tour leader. Do not touch restricted plants or greenhouse controls. Allergy: Plant and pollen sensitivity note.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Correct setting choices	30
Use of tour evidence	25
Explanation	20
Layout efficiency	15
Teamwork	10
Total	**100**

ACTIVITY TTT-06 · BIOMIMICRY

Pinecone Weather Machine

Build a humidity sensor with no battery and no code.

Win condition: Most accurate, readable, and fast response after spray-and-dry trials.

THE SCIENCE

Hygromorphs. A pine cone opens in dry air and closes when it is humid. Its scales are built from two layers of dead tissue that swell by different amounts when they absorb moisture, so the scale bends. No motor, no power, just material that responds to water in the air.

Bilayer biomimicry. You can copy this with a paper-and-tape bilayer. One layer absorbs moisture and grows; the other does not. The mismatch makes your indicator curl or straighten. The goal is a readable, repeatable humidity signal.

BUILD AND RUN IT

01 **Observe a real cone.** Spray a pine cone and watch the scales close. Let it dry and watch them open.

02 **Build a bilayer indicator.** Tape a moisture-absorbing layer to a stiff backing so the strip bends when wet.

03 **Add a readable scale.** Mount the strip against a ruler or dial so the bend points to a number.

04 **Run spray-and-dry trials.** Spray, time the response, dry, repeat. Record how far and how fast it moves.

05 **Redesign for speed.** Adjust layer thickness or length to respond faster and more clearly.

Safety: Spray water only at the one test spot and wipe up right away so the floor stays dry and no one slips. Inspect cones for sap, insects, and sharp broken scales. Handle skewer points carefully. Allergy: if you are sensitive to pine, use the paper-only bilayer instead of handling a cone. No goggles or gloves are needed.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Response accuracy	35
Response speed	20
Readable output scale	20
Biomimicry explanation	15
Redesign quality	10
Total	**100**

ACTIVITY TTT-07 · ECOLOGY

Pollinator Network Draft and Build

Draft a pollinator habitat that works all season, not just on the prettiest day.

Win condition: Highest pollinator support across spring, summer, and fall using native and clumping logic.

THE SCIENCE

Networks, not single plants. Pollinators need food across the whole season, not just one bloom. A good habitat is a network: plants that flower in spring, summer, and fall, matched to the pollinators that visit them. Gaps in the calendar starve the system.

Native and clumping logic. Native plants tend to support local pollinators best, and planting in clumps helps insects find and work them efficiently. You draft plant and pollinator cards, then place them on a grid under these constraints.

BUILD AND RUN IT

01 **Draft your cards.** Pick plant and pollinator cards. Note bloom season and which pollinators each plant supports (most flowers feed several kinds).

02 **Cover every season.** Lay out plants so something blooms in spring, summer, and fall. Use season tokens to check coverage.

03 **Apply native and clumping rules.** Favor natives and group same plants in clumps so pollinators forage efficiently.

04 **Connect the network.** Make sure each pollinator has food across its active months, not just one.

05 **Defend the design.** Explain how your grid supports the most pollinators across the whole season.

Safety: No live insects required. Students only handle cards and grid materials. Allergy: None.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Seasonal coverage	30
Pollinator fit	25
Native and clumping logic	20
Layout clarity	15
Defense	10
Total	**100**

Arboretum Eco-Quest

Read the living collection and solve the checkpoints before time runs out.

Win condition: Most accurate checkpoint solutions with safe field conduct.

THE SCIENCE

Observation as evidence. Identifying a tree is detective work: leaf shape, bark texture, branching pattern, and seeds are all clues. A dichotomous key turns those clues into a series of either-or choices that lead to a name.

Ecosystems in place. The arboretum is a living classroom of ecosystems. Each checkpoint asks you to observe carefully, classify what you see, and record the evidence that justifies your answer.

BUILD AND RUN IT

01 **Get the map and rules.** Each team gets a route map and the field-conduct rules. Stay on paths, stay together.

02 **Solve each checkpoint.** Use leaf, bark, and seed clues with the key to answer each station.

03 **Collect evidence tokens.** Record the specific clue that justifies each answer and collect the token.

04 **Keep an efficient pace.** Order your route to avoid backtracking; pace counts but accuracy counts more.

05 **Check your answers.** Before returning, confirm each answer cites real evidence, not a guess.

Safety: Stay on route. No off-trail shortcuts. Buddy system and hydration check. Allergy: Pollen and insect-sting awareness.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Checkpoint accuracy	50
Evidence quality	20
Pace	20
Teamwork and conduct	10
Total	**100**

ACTIVITY TTT-09 · ECOLOGY

Minecraft Tree World Resilience Cup

Design a landscape that survives stress because of how it is built, not luck.

Win condition: Best climate-resilient design with accurate nature-based strategies.

THE SCIENCE

Resilience by design. A resilient landscape keeps working after a shock: a storm, a drought, a heat wave. Diversity, connected habitats, shade, and water management all add resilience. Monocultures and bare ground fail fast.

Systems thinking. In Minecraft Education or a paper grid, you build a landscape and justify each choice with a nature-based strategy. The win is the most resilient, biodiverse, realistic design, not the prettiest screenshot.

BUILD AND RUN IT

01 **Pick a stress to survive.** Choose the challenge your landscape must withstand: flood, drought, or heat.

02 **Add resilience features.** Build diversity, connected green space, shade, and water control into the design.

03 **Support biodiversity.** Include varied habitats so many species can live there, not just one.

04 **Keep it realistic.** Use strategies that would actually work outdoors, and label them.

05 **Present and defend.** Walk judges through how each feature adds resilience.

Safety: Standard device rules. Keep logins prepared. Use paper-grid fallback if devices fail. Allergy: None.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Resilience features	30
Biodiversity support	25
Realism	20
Explanation	15
Build polish	10
Total	**100**

ACTIVITY TTT-10 · DATA + CLIMATE

Tree Ring Climate Detective

Read the rings to reconstruct the climate events a tree lived through.

Win condition: Most accurate climate-event inferences with claim-evidence-reasoning.

THE SCIENCE

Rings as proxy data. A tree adds one ring per year. Wide rings mean a good growing season; narrow rings mean stress like drought, cold, or crowding. Because the tree cannot record numbers, the rings are a proxy: an indirect record of past climate.

Claim, evidence, reasoning. You infer events from ring patterns, then justify each call with the specific rings that support it. Scars, sudden narrowing, and runs of wide rings are your clues. Strong inferences cite the evidence.

BUILD AND RUN IT

01 **Learn the ring code.** Wide equals a favorable year, narrow a stress year (drought, cold, or crowding), scar a fire the tree survived. A ring is a proxy, not a thermometer.

02 **Read the sequence.** Work along the rings from center to bark, noting each pattern and the year it maps to.

03 **Infer the events.** Match patterns to likely events: drought, fire, a crowded stand, a recovery.

04 **Cite your evidence.** For each inference, point to the exact rings that justify it on a claim-evidence card.

05 **Synthesize the story.** Combine your inferences into one short climate history for the tree.

Safety: Sand wood cookies smooth if used. No cutting tools with students. Allergy: Wood dust sensitivity note if using real samples.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Correct inferences	40
Evidence justification	20
Speed	20
Final synthesis	20
Total	**100**

Lotus Leaf Surface Sprint

Build a surface that sheds water and dirt the way a lotus leaf does.

Win condition: Best clean runoff and least residue after one redesign.

THE SCIENCE

Hydrophobic micro-texture. A lotus leaf is covered in tiny bumps coated in wax. Water cannot settle into the texture, so droplets bead up into near-spheres and roll off, carrying dirt with them. The effect comes from roughness plus a water-repelling coating, not from being smooth.

Roughness plus coating. You texture and coat safe surfaces, then race droplets down a standard ramp. The best surface sheds water fast and leaves the least residue. Copying the leaf on purpose is biomimicry in materials engineering.

BUILD AND RUN IT

01 **Compare bare surfaces.** Drop water on smooth and rough surfaces. Watch which makes the droplet bead up.

02 **Texture and coat.** Add micro-texture and a water-repelling layer to your test surface.

03 **Add a dirt challenge.** Sprinkle pepper or cocoa as fake dirt. A good surface carries it off with the droplet.

04 **Race down the ramp.** Run droplets down the standard ramp. Time the runoff and check the residue left behind.

05 **Redesign for less residue.** Adjust texture or coating to shed cleaner, then re-run.

Safety: Keep floors dry. Use washable residue. No chemical coatings beyond classroom-safe materials. Allergy: Cocoa allergy substitute: pepper or washable soil.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Fast clean runoff	35
Least residue	25
Surface design explanation	20
Redesign gain	10
Craftsmanship	10
Total	**100**

ACTIVITY TTT-12 · FIELD SCIENCE

Leaf Stomata Microscope Detective

Count the breathing pores and rank leaves by how they manage water.

Win condition: Most accurate detective ranking with clean data and strong evidence.

THE SCIENCE

Stomata: pores for gas exchange. Leaves breathe through tiny pores called stomata. They open to take in carbon dioxide and let oxygen out, but open stomata also lose water. The number and spacing of stomata reflect how a plant balances feeding itself against drying out.

Sampling and counting. You make a safe leaf peel or use prepared slides, count stomata in several fields of view, and average. Comparing counts across leaves lets you rank them by water strategy with real data.

BUILD AND RUN IT

01 **Make a safe peel.** Paint clear polish on the leaf underside, let it dry, and lift the print with clear tape. Or use a prepared slide.

02 **Focus the field.** Place the peel under the microscope and focus until the stomata are sharp.

03 **Count several fields.** Count stomata in three or more fields of view and average to reduce error.

04 **Compare leaves.** Repeat for different leaves and build a team data table of counts per field.

05 **Rank by water strategy.** Use your counts to rank leaves from water-saving to water-spending, citing the data.

Safety: Use polish only at ventilated instructor station. No chromatography, no solvents, no tie-dye. Allergy: Plant sensitivity note. Prepared slides available.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Slide or peel quality	25
Counting accuracy	30
Ranking evidence	25
Team data table	10
Cleanup and care	10
Total	**100**

BACKUP ACTIVITIES

If weather or materials change

ACTIVITY TTB-01 · FIELD SCIENCE

Root Grip Erosion Rescue

Can roots hold a slope together during a storm?

Win condition: Most soil retained after standard pour.

THE SCIENCE

Roots anchor soil. Bare soil on a slope washes away fast when rain hits it. Roots grip the soil and hold it in place, and a denser, deeper root network holds better. This is why planting vegetation is a real tool for stopping erosion.

Slope and runoff. Steeper slopes and bare surfaces shed water and soil quickly. You build a mini slope with root-like anchors, pour a standard amount of water, and measure how much soil stays put.

BUILD AND RUN IT

01 **Build a bare slope.** Pack soil into a tray at a set angle with no roots. This is your control.

02 **Add root anchors.** Push string or yarn roots into a second slope to mimic a planted hillside.

03 **Pour the standard storm.** Pour the same measured water on each slope from the same height.

04 **Measure soil retained.** Compare how much soil washed out of each tray. Record the difference.

05 **Redesign the root network.** Add more or deeper roots and re-test to retain more soil.

Safety: Use trays and wipe spills. Allergy: Dust/pollen sensitivity.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Soil retained	40
Runoff control	25
Design explanation	20
Cleanup	15
Total	**100**

ACTIVITY TTB-02 · FIELD SCIENCE

Tree Height Triangulation Shootout

Measure a tree without climbing it.

Win condition: Closest to instructor reference height.

THE SCIENCE

Angles give height. You can find a tree's height without touching the top. Sight the treetop through a paper clinometer to read an angle, measure your distance to the trunk, and the geometry of a right triangle gives the height. This is remote sensing with a protractor.

Accuracy from method. Small sighting errors grow over distance, so careful technique matters: level your eye, sight steadily, measure distance honestly. The team closest to the instructor's reference height wins.

BUILD AND RUN IT

01 **Build the clinometer.** Fold a paper clinometer with a weighted string plumb line to read angles.

02 **Pace a known distance.** Measure your straight-line distance from the trunk and record it.

03 **Sight the treetop.** Look along the clinometer at the very top and read the angle steadily.

04 **Compute the height.** Use the angle, your distance, and your eye height to estimate the tree height.

05 **Check against reference.** Compare to the instructor reference. The closest estimate wins.

Safety: Outdoor boundaries. Allergy: Pollen sensitivity.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Accuracy	50
Method clarity	20
Speed	20
Teamwork	10
Total	**100**

ACTIVITY TTB-03 · DATA + CLIMATE

Urban Heat Shade Dash

Find the coolest route through a hot campus.

Win condition: Best data-backed cool route.

THE SCIENCE

Urban heat and shade. Paved, sunlit surfaces get much hotter than shaded or planted ones. Shade from trees and buildings can drop surface temperature sharply over a few steps. Mapping these differences reveals cooler paths through a hot space.

Data-backed routing. You measure shaded and sunny surfaces, then design a cool corridor: a route that keeps people on the coolest surfaces. The best route is justified by your own temperature data.

BUILD AND RUN IT

01 **Measure sun vs shade.** Record surface temperature in matched sunny and shaded spots.

02 **Map the hot spots.** Mark the hottest and coolest surfaces on your map.

03 **Design a cool corridor.** Plot a walking route that stays on the coolest surfaces.

04 **Justify with data.** Show the temperatures that make your route cooler than the direct path.

05 **Communicate the plan.** Present the route clearly so someone could follow it.

Safety: Check heat index. Walk, do not run. Allergy: Pollen sensitivity.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Cool route design	35
Data quality	25
Reasoning	20
Communication	20
Total	**100**

ACTIVITY TTB-04 · FIELD SCIENCE

Photosynthesis Float-Off Playoffs

Make leaf disks float by producing oxygen.

Win condition: Fastest half-float time with controlled variables.

THE SCIENCE

Photosynthesis makes oxygen. Leaf disks sink when the air is drawn out of them. In light, with carbon dioxide available from a baking-soda solution, the leaf photosynthesizes and produces oxygen. The oxygen collects inside the disk until it floats. Floating is direct evidence of photosynthesis.

Controlled variables. Light level, carbon-dioxide concentration, and temperature all change the rate. You run a controlled assay, change one variable, and time how long until half the disks float.

BUILD AND RUN IT

01 **Cut and sink the disks.** Punch leaf disks, then use a needle-free syringe to pull the air out so they sink.

02 **Set up the solution.** Place disks in baking-soda solution to supply carbon dioxide.

03 **Add light.** Put the cup under a light source to drive photosynthesis.

04 **Time the half-float.** Record the time until half the disks rise to the surface.

05 **Change one variable.** Adjust light or carbon-dioxide level, hold the rest constant, and compare rates.

Safety: No tasting. Handle syringes as lab tools. Allergy: Spinach handling note.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Performance	35
Controlled-variable design	25
Data table	20
Explanation	10
Cleanup	10
Total	**100**